

Sine function



1 Consider the function $y = \sin x$.

a Complete the table of values giving answers in exact form:

x	0°	30°	90°	150°	180°	210°	270°	330°	360°
$\sin x$									

b Sketch the graph of $y = \sin x$ for $0 \leq x \leq 360$.

c State the sign of $\sin 324^\circ$.

d Which quadrant would the angle 324° lie?

e Describe the trend of the graph of $y = \sin x$.

f State the period of $y = \sin x$.

g State the y -intercept.

h State the maximum y -value.

i State the minimum y -value.

j State the range of the function.

2 Consider the function $y = \sin x$.

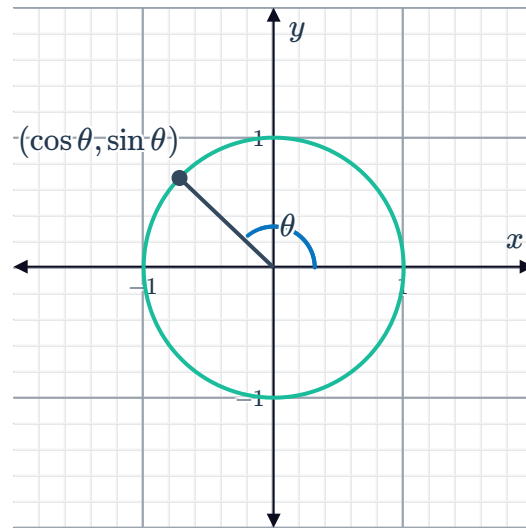
a Complete the table of values giving answers in exact form:

x	0	$\frac{\pi}{6}$	$\frac{\pi}{2}$	$\frac{5\pi}{6}$	π	$\frac{7\pi}{6}$	$\frac{3\pi}{2}$	$\frac{11\pi}{6}$	2π
$\sin x$									

b Sketch the graph of $y = \sin x$ for $0 \leq x \leq 2\pi$.

c State the period of $y = \sin x$.

- 3 Given the unit circle, explain why the following statements are true for the graph of $y = \sin \theta$:



- a The values of $y = \sin \theta$ lie in the range $-1 \leq y \leq 1$.
- b The graph of $y = \sin \theta$ repeats after every 360° .

- 4 Determine whether the following statements are true of the curve $y = \sin x$:

- a The graph of $y = \sin x$ is cyclic.
- b As x approaches infinity, the height of the graph approaches infinity.
- c The graph of $y = \sin x$ is increasing between $x = -360^\circ$ and $x = -270^\circ$.
- d The graph of $y = \sin x$ is symmetric about the line $x = 0$.
- e The graph of $y = \sin x$ is symmetric with respect to the origin.
- f The y -values of the graph repeat after a period of 180° .
- g As x approaches infinity, the graph of $y = \sin x$ stays between $y = -1$ and $y = 1$.

- 5 Consider the function $y = \sin x$.

- a Describe whether the graph is increasing or decreasing in the following intervals:

- | | |
|--------------------------------|----------------------------------|
| i $-270^\circ < x < -90^\circ$ | ii $-450^\circ < x < -270^\circ$ |
| iii $90^\circ < x < 270^\circ$ | iv $-90^\circ < x < 90^\circ$ |

- b Find the x -intercept in the following intervals:

- | | |
|-----------------------------|-------------------------------|
| i $0^\circ < x < 360^\circ$ | ii $-360^\circ < x < 0^\circ$ |
|-----------------------------|-------------------------------|

6 Consider the function $y = \sin x$.



a Describe whether the graph is increasing or decreasing in the following intervals:

i $\frac{\pi}{2} < x < \frac{3\pi}{2}$

ii $\frac{3\pi}{2} < x < \frac{5\pi}{2}$

iii $-\frac{3\pi}{2} < x < -\frac{\pi}{2}$

iv $-\frac{5\pi}{2} < x < -\frac{3\pi}{2}$

b Find the x -intercept in the following intervals:

i $0 < x < 2\pi$

ii $-2\pi < x < 0$

7 The number of people visiting Times Square can be modeled by the function

$$P(t) = 24\,000 \sin \frac{\pi}{12}t + 26\,000$$

where $P(t)$ represents the number of people visiting after t hours.

a State the maximum value of this function over one full cycle.

b State the meaning of the answer from part (a) in context.

Cosine function

8 Consider the function $y = \cos x$.

a Complete the table of values giving answers in exact form:

x	0°	60°	90°	120°	180°	240°	270°	300°	360°
$\cos x$									

b Sketch the graph of $y = \cos x$ for $0 \leq x \leq 360$.

c State the sign of $\cos 340^\circ$.

d State the quadrant where an angle with measure 340° lie.

e State the y -intercept.

f State the maximum y -value.

g State the minimum y -value.

h State the domain of the function.

9 Consider the function $y = \cos x$.



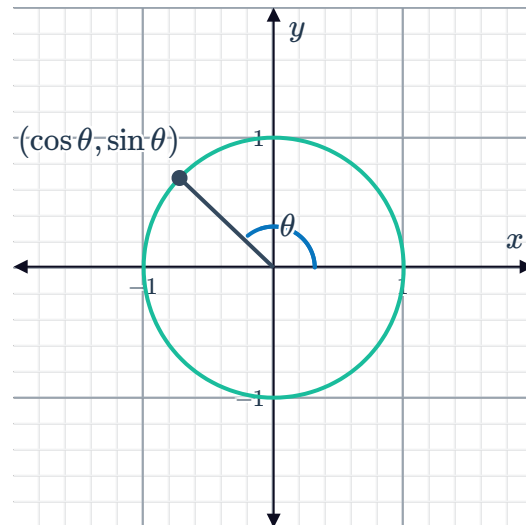
a Complete the table of values giving answers in exact form:

x	0	$\frac{\pi}{3}$	$\frac{\pi}{2}$	$\frac{2\pi}{3}$	π	$\frac{4\pi}{3}$	$\frac{3\pi}{2}$	$\frac{5\pi}{3}$	2π
$\cos x$									

b Sketch the graph of $y = \cos x$ for $0 \leq x \leq 2\pi$.

10 Given the unit circle, explain why the following statements are true for the graph of $y = \cos \theta$:

- a The values of $y = \cos \theta$ lie in the range $-1 \leq y \leq 1$.
- b The graph of $y = \cos \theta$ repeats after every 360° .



11 State whether the following statements are true of the graph of $y = \cos x$:

- a The graph of $y = \cos x$ is cyclic.
- b As x approaches infinity, the height of the graph approaches infinity.
- c The graph of $y = \cos x$ is increasing between $x = 90^\circ$ and $x = 180^\circ$.
- d The graph of $y = \cos x$ is symmetric about the line $x = 0$.
- e The graph of $y = \cos x$ is symmetric with respect to the origin.
- f The y -values of the graph repeat after a period of 180° .
- g As x approaches infinity, the graph of $y = \cos x$ stays between $y = -1$ and $y = 1$.

12 Consider the function $y = \cos x$.



- a If one cycle of the graph of $y = \cos x$ starts at $x = -90^\circ$, where does the next cycle start?
- b Describe whether the graph of $y = \cos x$ is increasing or decreasing in the following intervals:
 - i $-180^\circ < x < 0^\circ$
 - ii $-360^\circ < x < -180^\circ$
 - iii $0^\circ < x < 180^\circ$
 - iv $180^\circ < x < 360^\circ$
- c Find the x -value of the x -intercepts in the following intervals:
 - i $0^\circ < x < 360^\circ$
 - ii $-360^\circ < x < 0^\circ$

13 Consider the function $y = \cos x$.

- a If one cycle of the graph of $y = \cos x$ starts at $x = -\frac{\pi}{2}$, where does the next cycle start?
- b Describe whether the graph of $y = \cos x$ is increasing or decreasing in the following intervals:
 - i $\pi < x < 2\pi$
 - ii $-2\pi < x < -\pi$
 - iii $0 < x < \pi$
 - iv $-\pi < x < 0$
- c Find the x -value of the x -intercepts in the following intervals:
 - i $0 < x < 2\pi$
 - ii $-2\pi < x < 0$

14 The average monthly temperature of a city can be modeled by a cosine graph. Christa has been living in Brooklyn, New York, where the average annual temperature is 55°F . She would like to move, and live in a location where the average annual temperature is 72°F . When examining the graphs of the average monthly temperatures for various locations, determine whether Christa should focus on each of the following:

- a Amplitude
- b Midline
- c Horizontal shift
- d Period



Tangent function

15 Consider the equation $y = \tan x$.

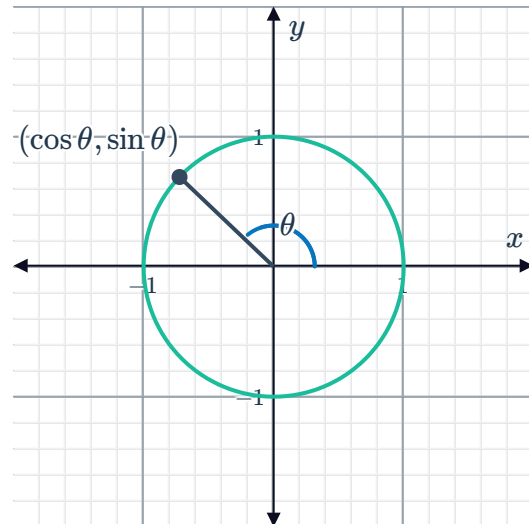
- a Complete the table of values, giving answers in exact form. Write '-' if the value is undefined:

x	0°	60°	90°	120°	135°	180°	225°	270°	300°	360°
$\tan x$										

- b Sketch the graph of $y = \tan x$ for $0 \leq x \leq 360$.
 c State the y -intercept.
 d State the sign of $\tan 345^\circ$.
 e State the quadrant where an angle with measure 345° lie.

16 Given the unit circle below, state whether the following statements are true of the graph of $y = \tan x$:

- a The graph of $y = \tan x$ repeats in regular intervals since the values of $\sin x$ and $\cos x$ repeat in regular intervals.
 b The graph of $y = \tan x$ is defined for all values of x .
 c Since the radius of the circle is one unit, the value of the function $y = \tan x$ lies in the region $-1 \leq y \leq 1$.
 d The range of values of function $y = \tan x$ is $-\infty < y < \infty$.



17 Consider $y = \tan x$ for $-360^\circ \leq x \leq 360^\circ$.

- a Determine the period of $y = \tan x$ in degrees.
 b Determine the range of $y = \tan x$.
 c As x increases, what will be the next asymptote of the graph after $x = 450^\circ$?

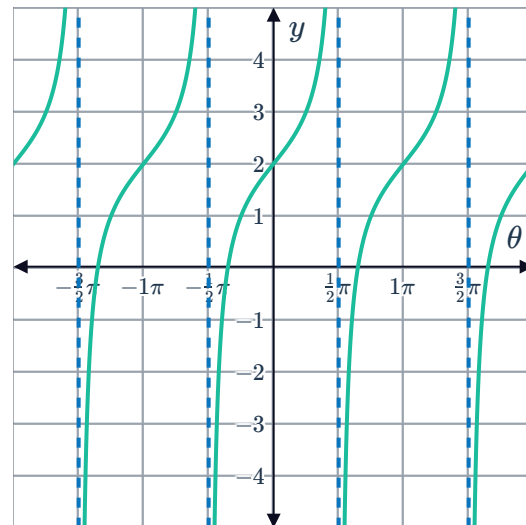
18 Consider $y = \tan x$ for $-2\pi \leq x \leq 2\pi$.



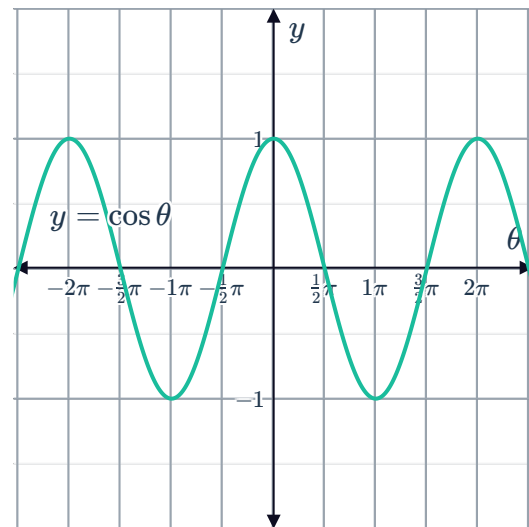
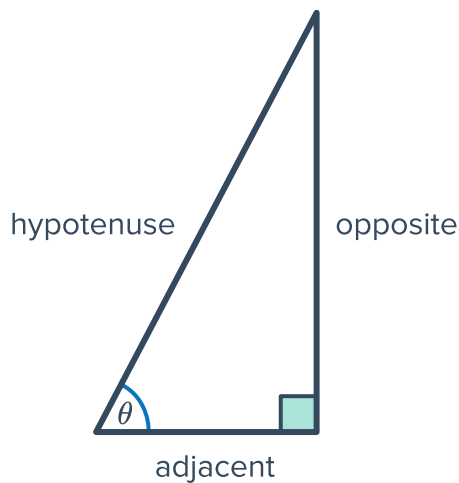
- a State the y -intercept of the graph.
- b State the period of the function in radians.
- c State the equations of the vertical asymptotes on the domain $0 \leq x \leq 2\pi$.
- d Does the graph of $y = \tan x$ increase or decrease between any two successive vertical asymptotes?
- e If $x > 0$, find the least value of x for which $\tan x = 0$.

19 Consider the graph of $y = \tan x + 2$.

- a How long is one cycle of the graph?
- b State the x -values for which the equation $\tan x + 2 = 2$, from $x = -2\pi$ to $x = 2\pi$ inclusive.
- c State the first positive x -value for which $\tan x + 2 = 3$.
- d Using the period of the graph, for what other values of x between $x = -2\pi$ and $x = 2\pi$ does $\tan x + 2 = 3$?
- e For what values of x between $x = -2\pi$ and $x = 2\pi$ does $\tan x + 2 = 1$?



20 Consider the right triangle containing angle θ and the graph of $y = \cos \theta$:



- As angle θ increases from 0 to $\frac{\pi}{2}$, explain what happens to the value of the opposite side in the triangle.
- Now, explain what happens to the value of $\tan \theta$ as angle θ increases from 0 to $\frac{\pi}{2}$, given that $\tan \theta$ is defined as $\frac{\text{opposite}}{\text{adjacent}}$.
- Express $\tan \theta$ in terms of $\sin \theta$ and $\cos \theta$.
- Determine the values of θ for which $\cos \theta = 0$, given $-2\pi \leq \theta \leq 2\pi$.
- Now, state the values of θ between -2π and 2π for which $\tan \theta$ is undefined.
- Complete the tables below:

θ	-2π	$-\frac{7\pi}{4}$	$-\frac{5\pi}{4}$	$-\pi$	$-\frac{3\pi}{4}$	$-\frac{\pi}{4}$
$\tan \theta$						

θ	0	$\frac{\pi}{4}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{7\pi}{4}$	2π
$\tan \theta$							

- Now, sketch the graph of $y = \tan \theta$ on the domain $-2\pi \leq \theta \leq 2\pi$.